

📈 AUDIO FEATURE ANALYSIS

How Did World War II Affect Musical Characteristics?

A quantitative investigation into musical trends, emotional valence, pace, and acoustic shifts using Spotify track metadata.



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Spotify Dataset



Valence
Mood Shift

Tempo
BPM Trends

Energy
Acoustics

Track Audio Features: Metadata, Rhythm & Acoustics

📁 Basic Metadata

- **Artists:** Creators & performers listed for the track.
- **Explicit:** Content rating flag (1 = Explicit, 0 = Clean).
- **Valence:** Musical positiveness (Happy vs. Melancholic).
- **Popularity:** Score from 0 – 100 based on total stream count and recent plays.
- **Mode:** Major or Minor Key. (1, 0).

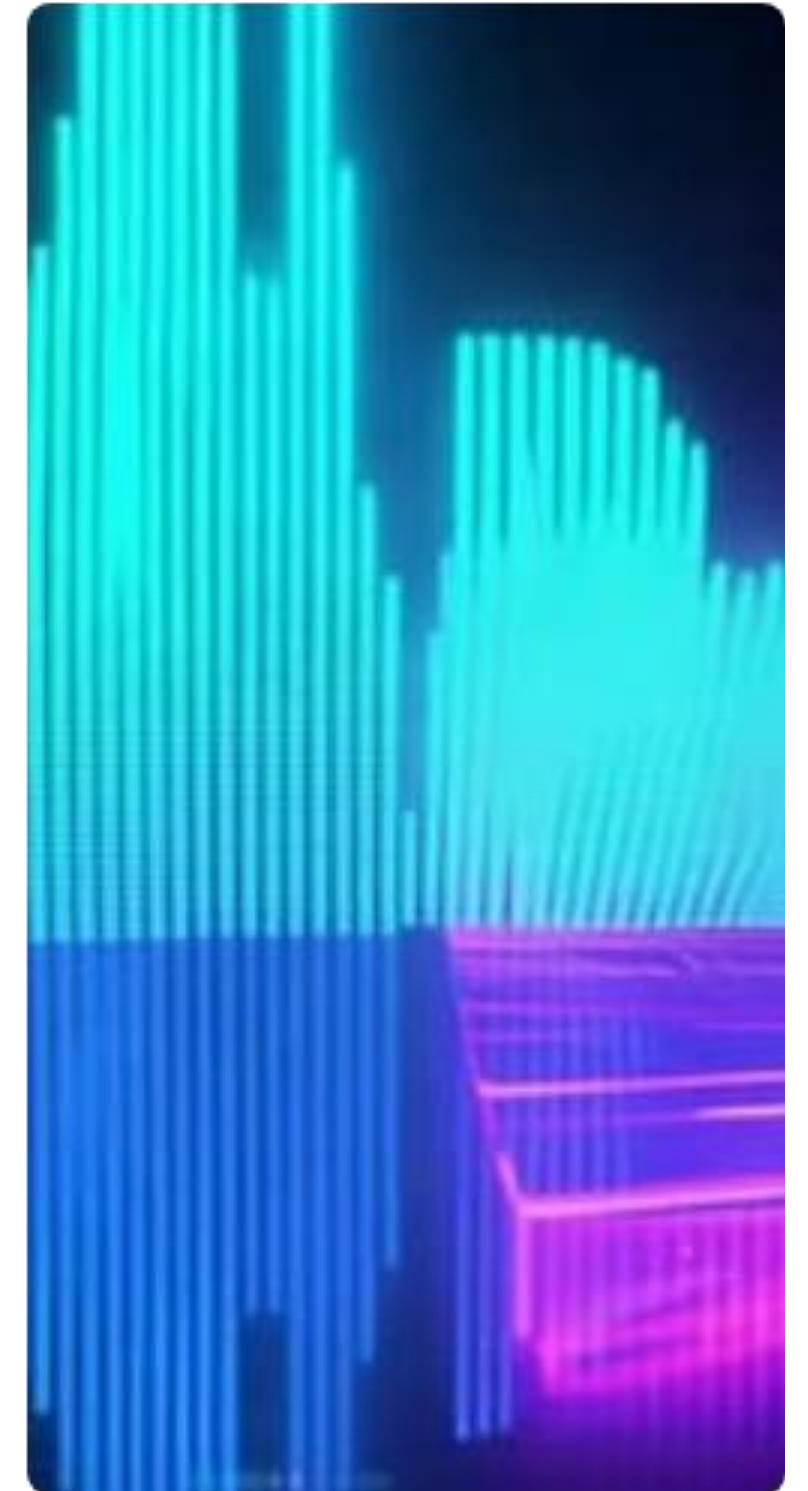
⚡ Rhythm & Pace

- **Danceability:** Suitability for dancing based on tempo stability and beat strength (0.0 – 1.0).
- **Energy:** Perceived intensity and activity level. Fast, loud, and noisy tracks score higher.
- **Tempo:** Overall speed or pace of the track measured in Beats Per Minute (BPM).

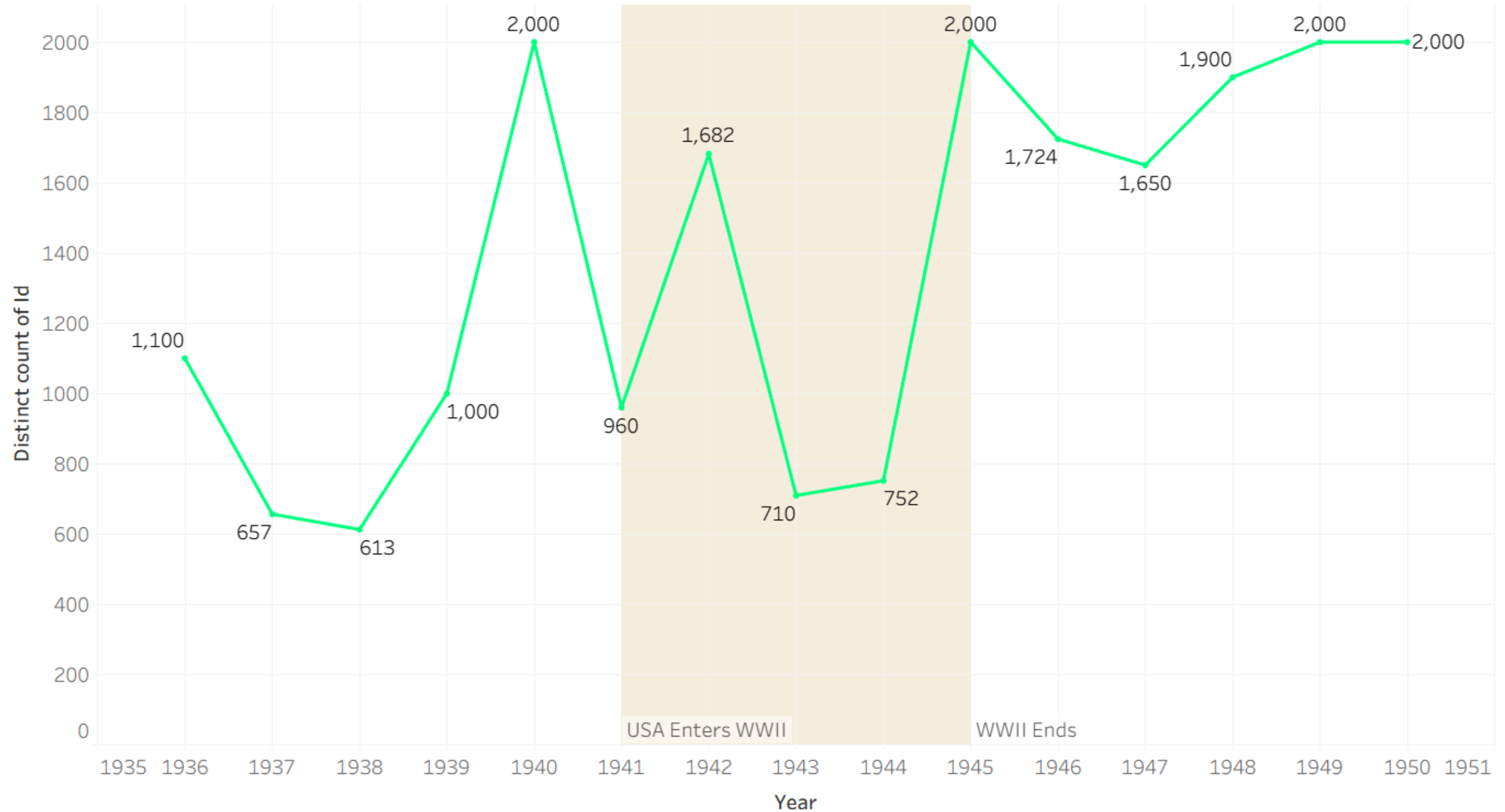
🔊 Acoustics & Vocals

- **Acousticness:** Presence of acoustic instruments vs digital amplification (0.0 – 1.0).
- **Instrumentalness:** Likelihood of no vocal content (High score = instrumental track).
- **Speechiness:** Presence of spoken words (podcasts, rap, spoken word).
- **Liveness:** Probability of live performance/audience presence (> 0.8).

🔊 Audio Signal



Count of songs



Song Concentration By Mode Per Year - Pre and Post WWII

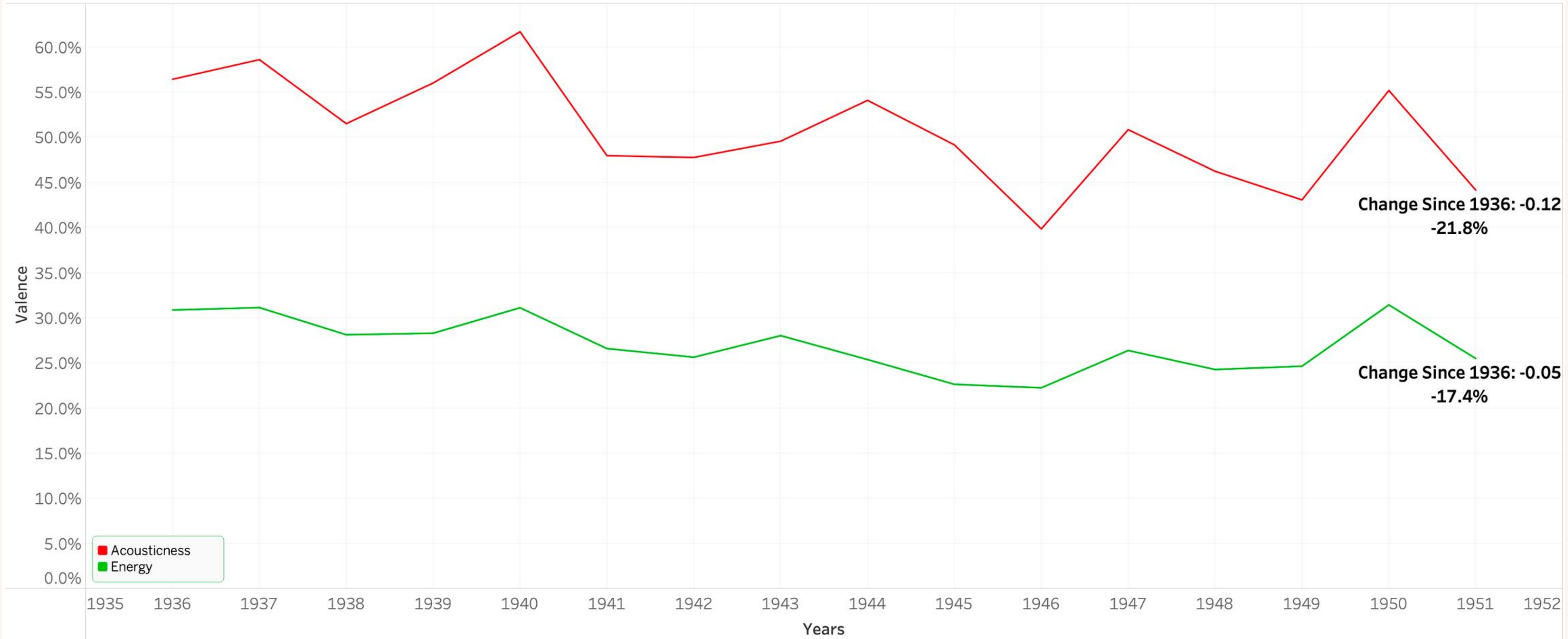
What is the concentration of songs by Mode? Major Key (1) Dominates.



Valence vs. Energy Over Years - Pre and Post WWII

Did WWII period shift the mood for songs? Yes. +Sad -Energetic.

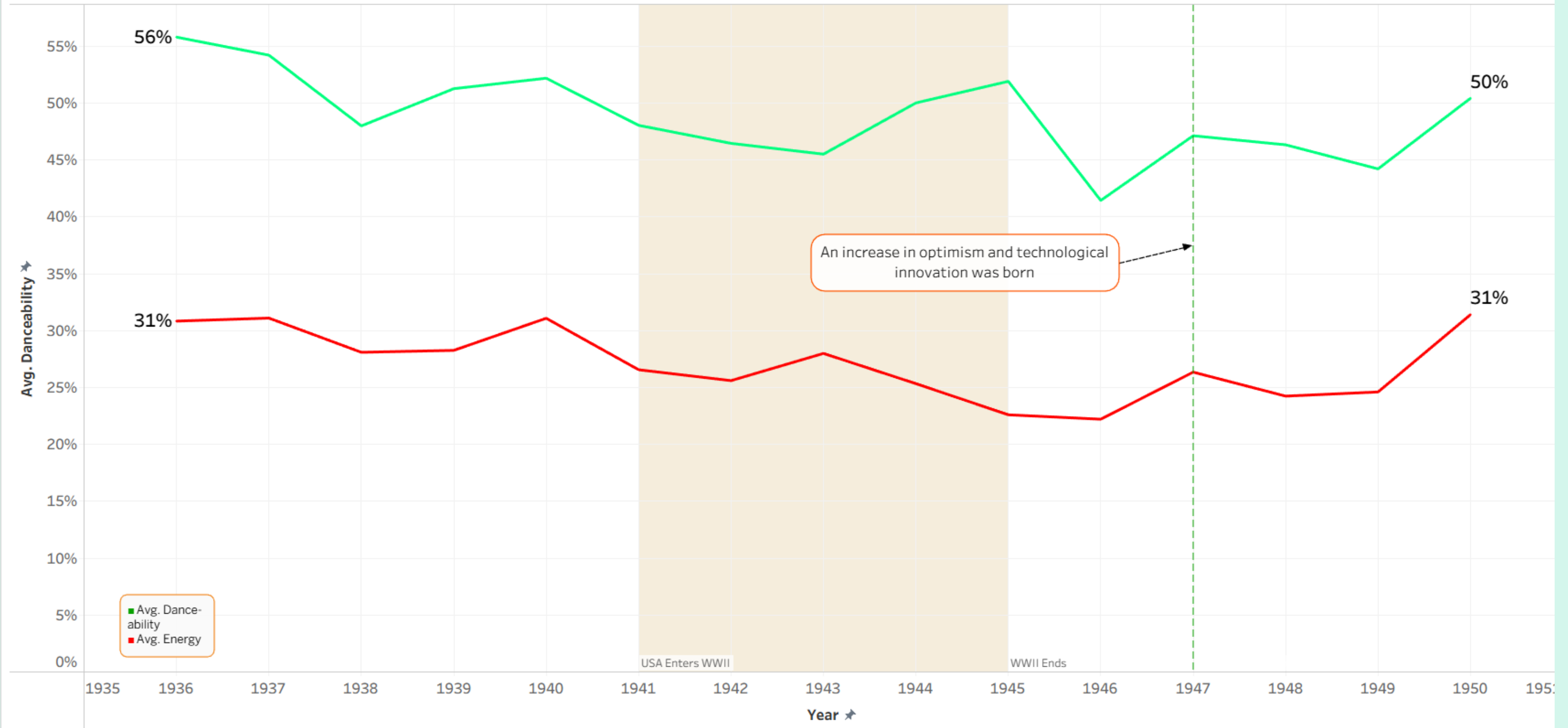
Valence vs Energy Over Years Pre and Post WWII



Danceability vs. Energy

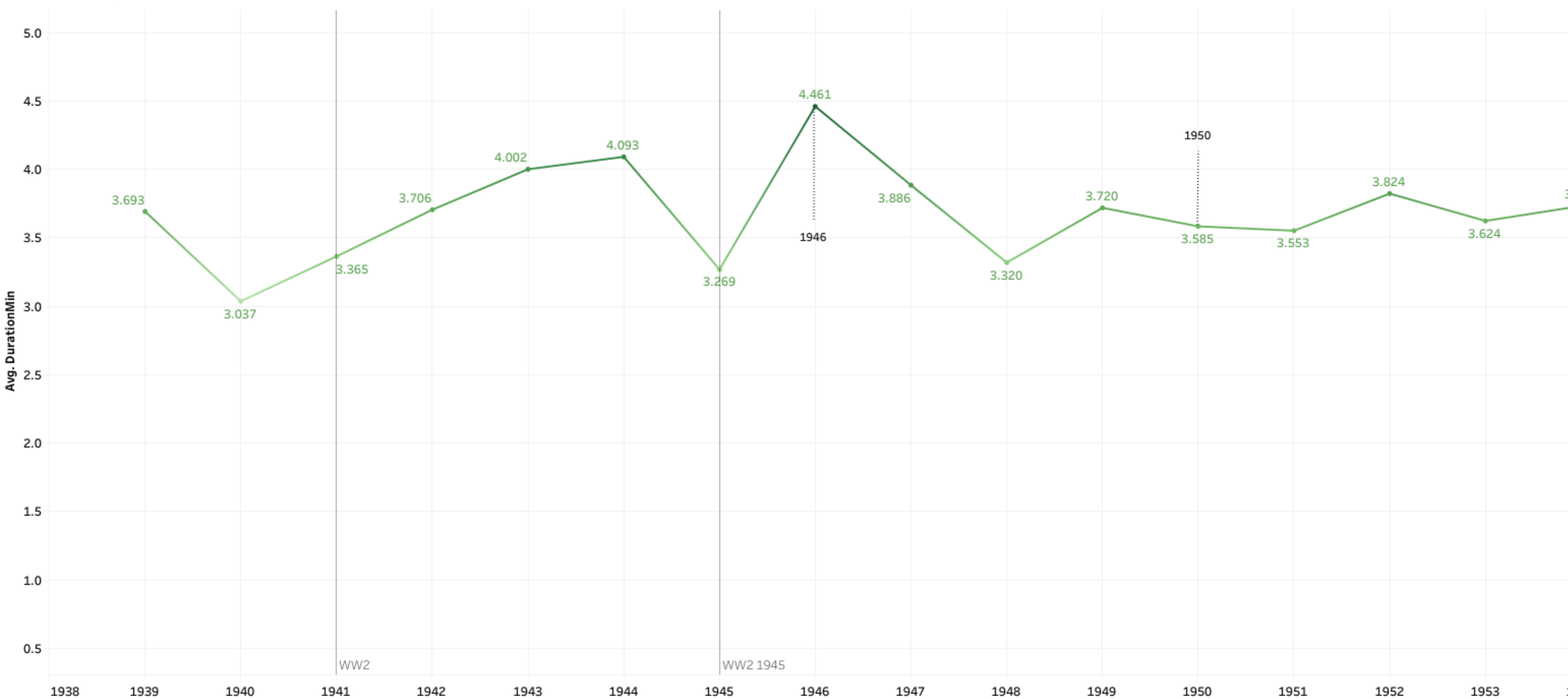
Why is there a shift right after WWII?

Danceability compared to energy



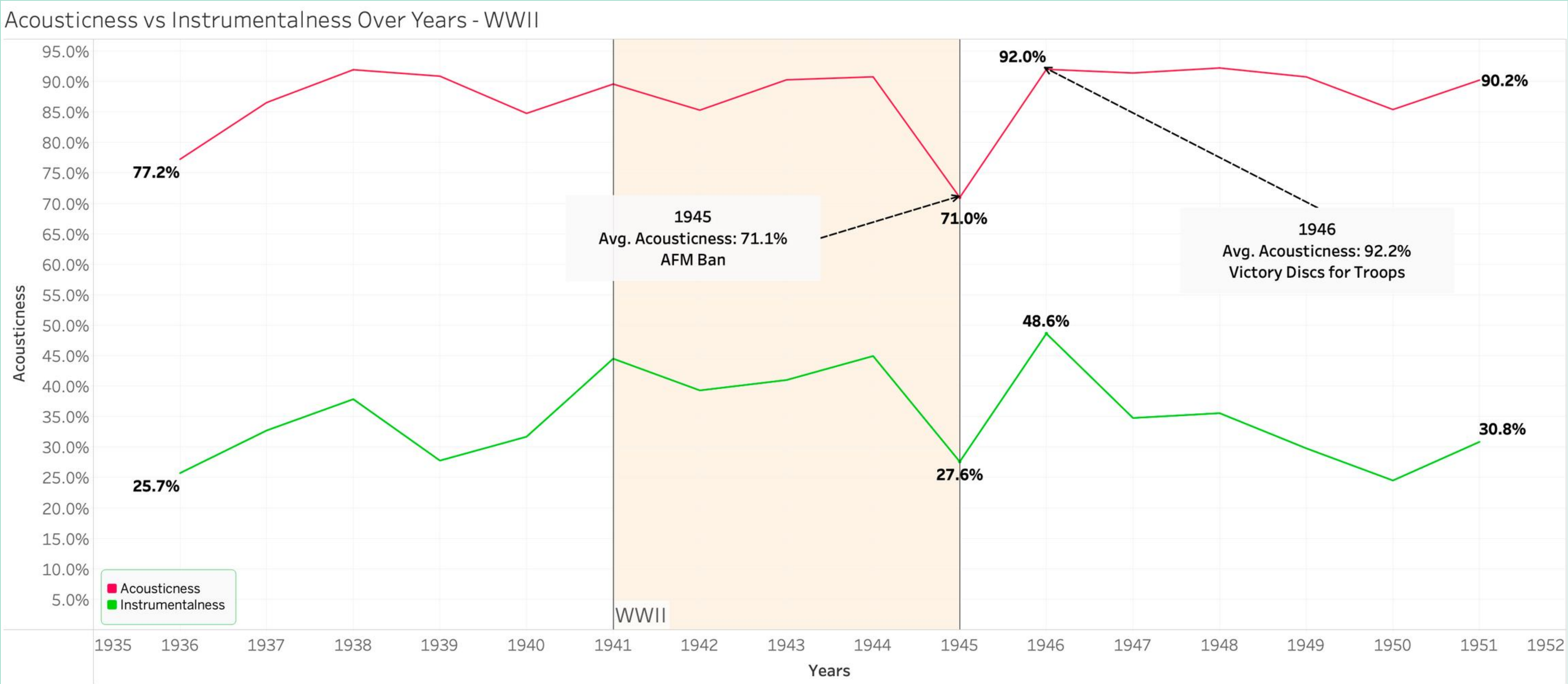
Changes of Duration During WW2

Duration Change in WWII



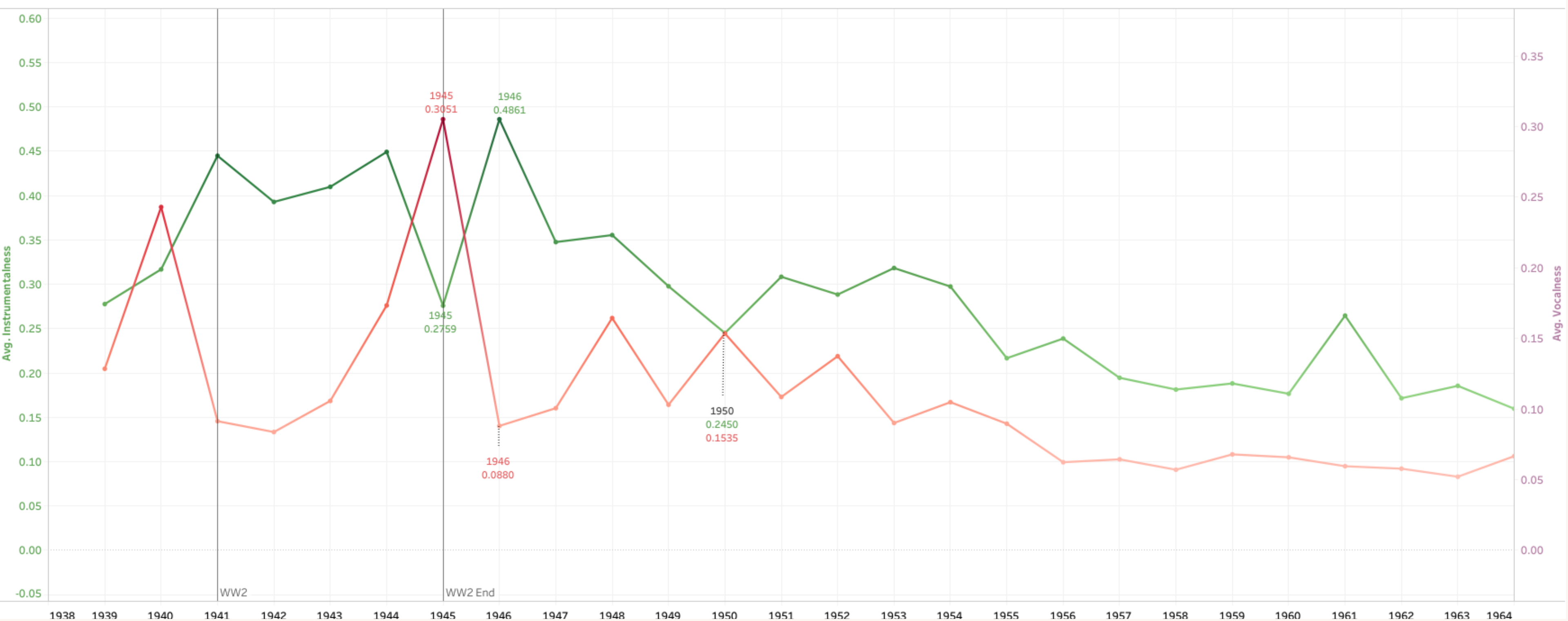
Acousticness vs. Instrumentalness WWII Period

What caused the massive acoustic spikes from 1945-1946? AFM Ban + Victory Discs.

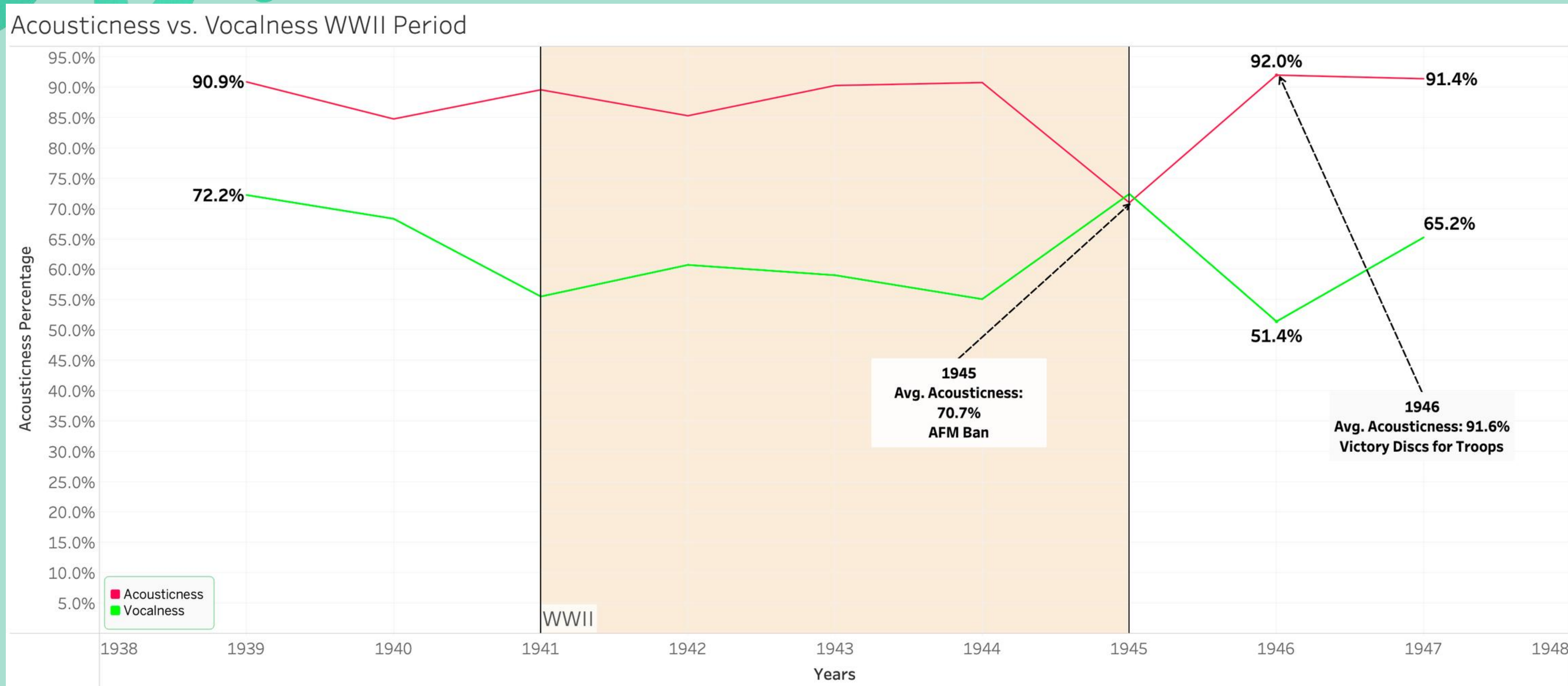


Shifts in Regards To Instrumentalness and Vocalness in WWII

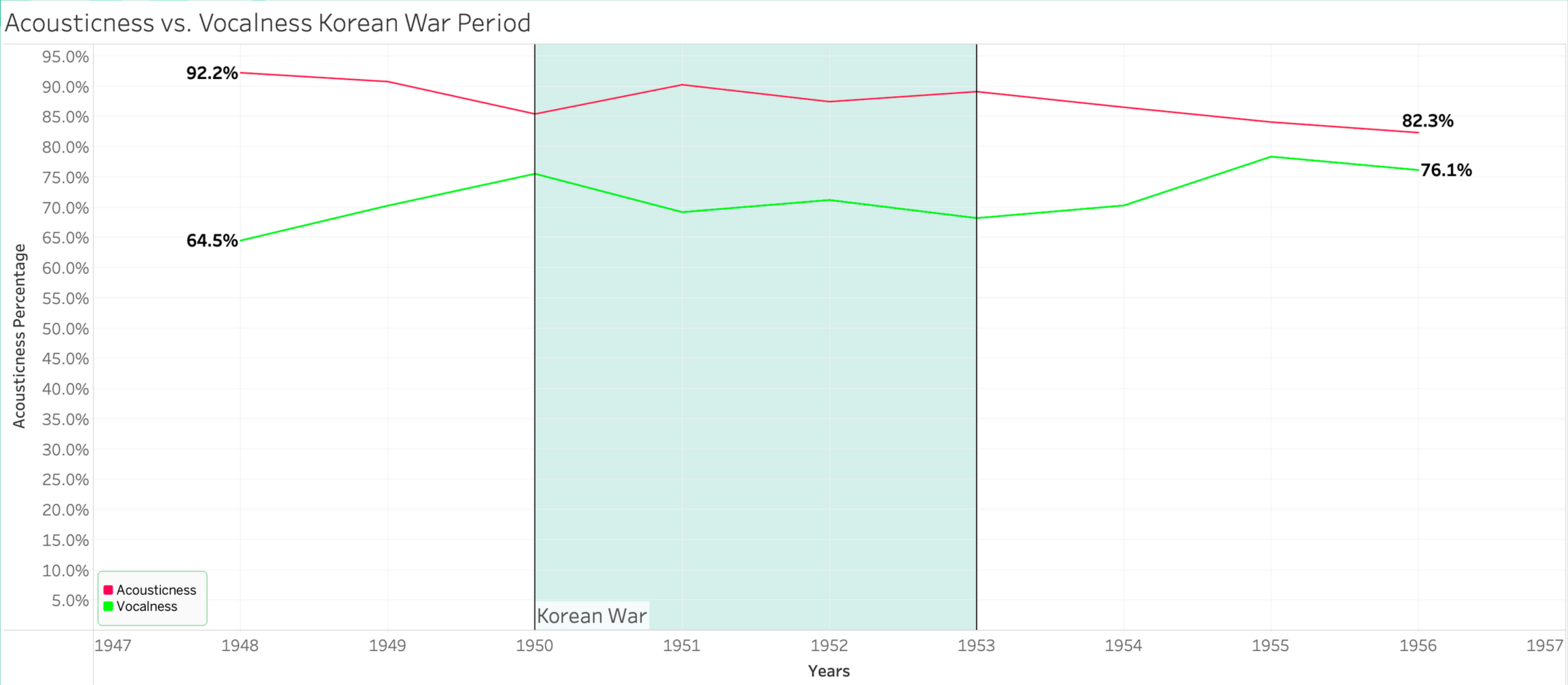
Instrumentalness & Vocalness



Acousticness vs. Vocalness Over - WWII

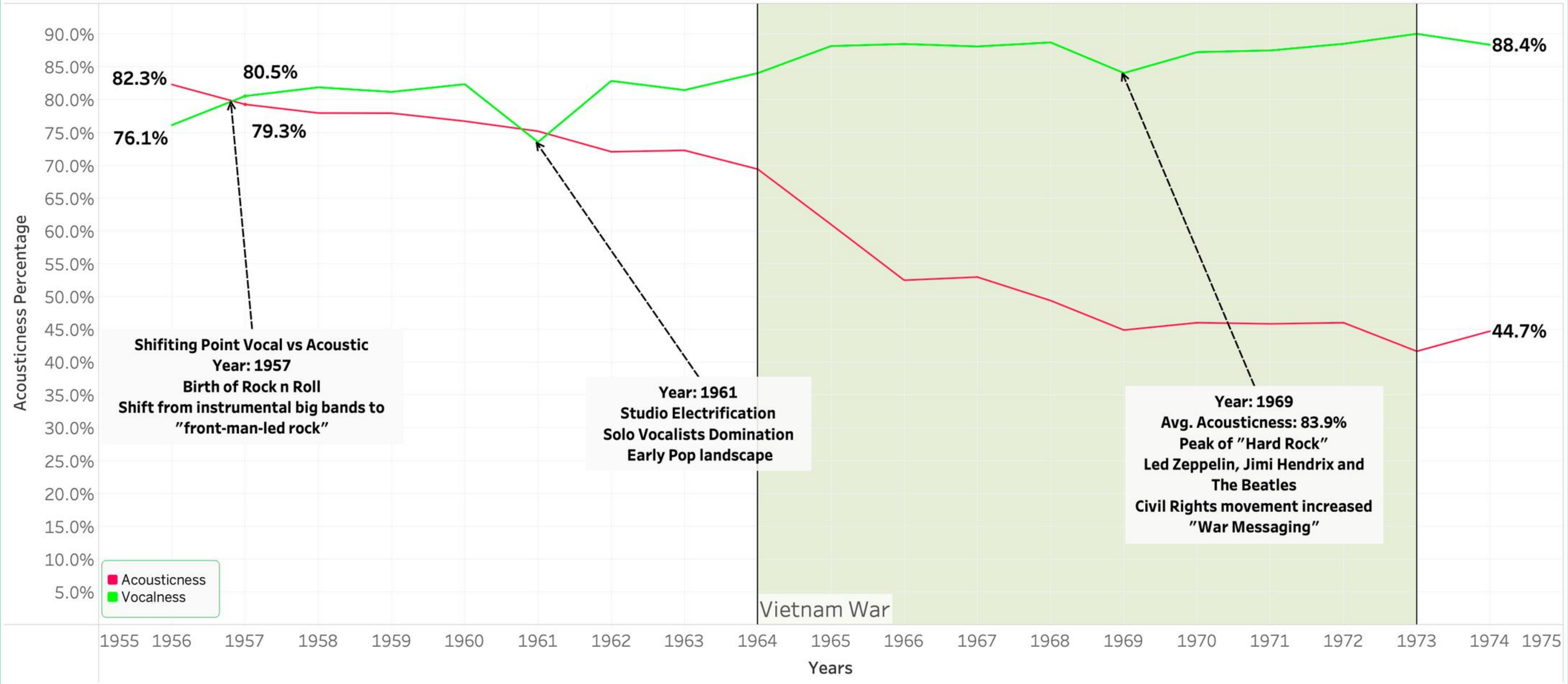


Acousticness vs. Vocalness Over - Korean War



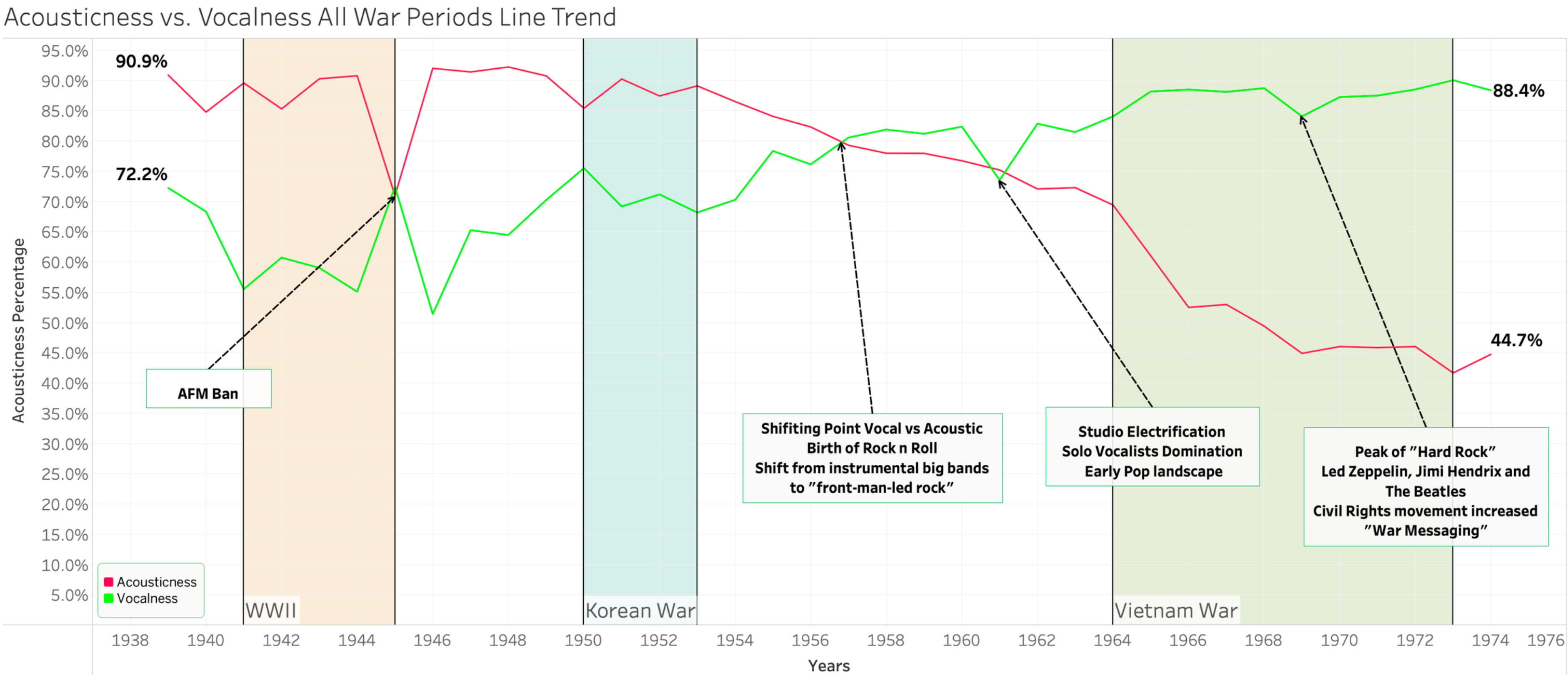
Acousticness vs. Vocalness Over - Vietnam War

Acousticness vs. Vocalness Vietnam War Period



Acousticness vs. Vocalness Over - Different Wars Perspective

Line Trend Behavior: Major shift from start to end.



Conclusion & Key Takeaways

Core Conclusion

WWII radically restructured modern music. It forced an immediate shift toward comforting, shorter formats before laying the technical foundation for the post-war vocal era and radio-standardized pop.

01

Valence & Energy Drop

Positivity fell by -22% and energy by -17.4% (1939–1945) due to a demand for comforting, melancholic tracks.

02

Technical Evolution

Post-war optimism and rapid studio electrification permanently transitioned the industry from big band instruments to amplified solo vocalists.

03

Long-Term Impact

Acousticness steadily declined to birth Rock n' Roll, and erratic war-era track lengths finally stabilized into modern, radio-friendly pop durations.